Memory Controllers for DDR Memories

Mentor: Mahesh A

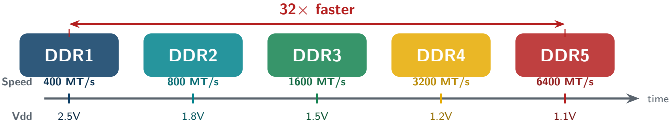
Pranav M PES2UG23EC076

Lalith PES2UG23EC074

Harshini S PES2UG23EC058

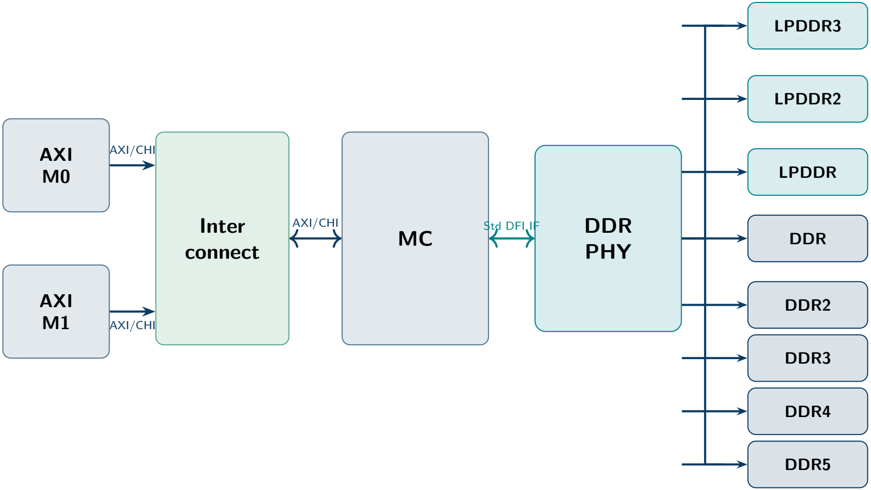
Recap: Memory Controller for DDR Memories

- Acts as a bridge between the SoC and DRAM
- Enforces 50+ timing constraints (tWTR, tRC, tFAW...)
- Handles row-buffer management and request scheduling
- Manages power modes: PASR, TCSR, Deep Power-Down
- AXI out-of-order IDs map naturally to DDR bank parallelism

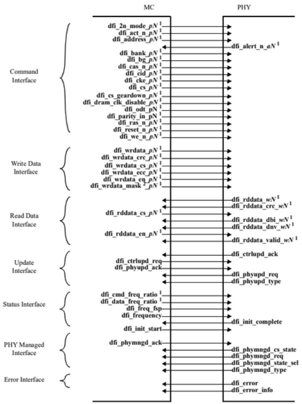


	DDR1	DDR2	DDR3	DDR4	DDR5
Prefetch	2n	4n	8n	8n	16n
Banks	4	4	8	16	32

System Block Diagram



DFI Interface: MC $\leftrightarrow$ PHY Signal Groups



Command Interface

- Address, bank, CS#, CKE, ODT, RAS#, CAS#, WE#
- MC $\rightarrow$ PHY; alert feedback PHY $\rightarrow$ MC

Write Data Interface

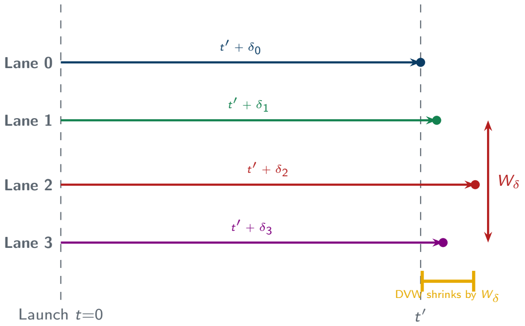
- `dfi_wrdata`, `dfi_wrdata_en`, `dfi_wrdata_mask`
- CRC and ECC sideband signals

Read Data Interface

- `dfi_rddata_en` MC $\rightarrow$ PHY (look-ahead)
- `dfi_rddata`, `dfi_rddata_valid` PHY $\rightarrow$ MC

Inter-Lane Skew: Off-Die Signal Integrity

- DDR transfers data in parallel across multiple DQ byte lanes
- Each lane follows a different PCB path, picking up delay δ_i
- Skew window: $W_\delta = \max(\delta_i) - \min(\delta_i)$
- DDR5 UI $\approx$ 156 ps; 20 ps skew eats >12% of it
- PHY training (Write Levelling, DQS Gating) corrects δ_i per lane



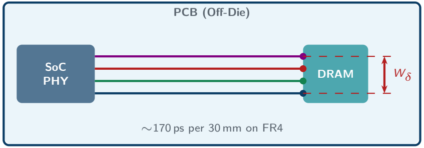
Off-Die vs On-Die Delay

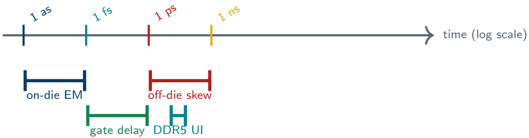
Off-Die (Picoseconds)

- Signals traverse PCB traces, vias
- FR4 velocity $\approx 0.6c$; $\sim$1 ps per 180 µm
- 30 mm trace $\approx$ 170 ps; W_δ typically 5–50 ps

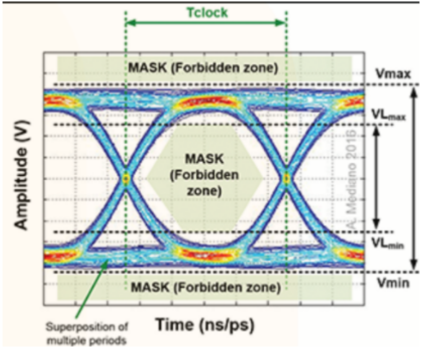
On-Die (Femto/Attoseconds)

- nm-scale on-chip metal interconnect
- 1 µm wire $\approx$ 3–10 as propagation
- Gate delay (3 nm): 1–5 ps; bottleneck is logic depth





Eye Diagram Fundamentals



Key Regions

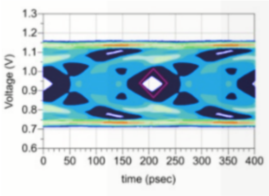
- **Eye Width:** horizontal opening between transition windows
- **Transition Window:** region where 0→1 or 1→0 switching occurs
- **Steady-State Window:** stable HIGH or LOW; sampled here
- **Unit Interval (UI):** one bit period; eye must open within it

Why it matters

- Receiver samples at eye centre for maximum margin
- Noise and jitter reduce eye opening
- At DDR5-6400: UI = 156 ps — any skew directly closes the eye

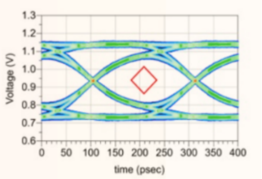
Eye Diagram: Trained vs Untrained

Untrained Eye



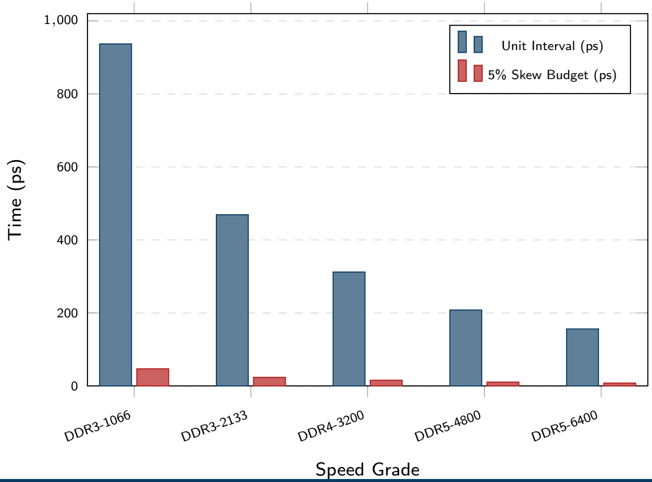
- Lane skew δ_i closes eye horizontally
- VREF offset closes eye vertically
- ISI and jitter blur transition edges
- High BER — unreliable data capture

Trained & Calibrated Eye



- Write Levelling corrects DQS-to-CK skew δ_i
- Read centering places sample point at eye centre
- ZQ calibration sets correct driver impedance
- 2D VREF scan optimises voltage margin M_V

Why Picoseconds Matter at Modern DDR Speeds



Speed	UI	5% Budget
DDR3-1066	937 ps	46.9 ps
DDR3-2133	469 ps	23.4 ps
DDR4-3200	312 ps	15.6 ps
DDR5-4800	208 ps	10.4 ps
DDR5-6400	156 ps	**7.8 ps**

- At DDR5-6400 the skew budget is just **7.8 ps**
- Uncompensated off-die skew can fill this entirely
- PHY training trims δ_i to $\ll$5 ps per lane

DFI: The Standard MC–PHY Interface

- **DFI** is the industry-standard interface between MC and DDR PHY
- Decouples MC from PHY implementation details
- Covers command, data, training and power signals
- DFI 5.1: PHY-independent boot, expanded freq range, enhanced calibration feedback

Ver	Key Addition	Support
1.0	Initial spec	DDR, DDR2
2.0	Training interface	DDR3, LPDDR2
3.0	Low-power handshake	DDR4, LPDDR4
4.0	DDR4/LPDDR4 paths	DDR4, LPDDR4
5.0	PHY-independent boot	DDR5, LPDDR5
5.1	**Full back-compat**	**All gens**

DFI 5.1 Key Features

Three pillars of DFI 5.1

1. **PHY-Independent Boot**
   MC initialises DRAM without knowing PHY internals

2. **Expanded Frequency Range**
   Covers low-power LPDDR through DDR5-6400

3. **Enhanced PHY↔MC Interaction**
   Real-time calibration feedback and margining

Backward compatibility

- Supports DDR through DDR5 and LPDDR through LPDDR5
- One DFI 5.1 MC paired with generation-specific PHY covers every standard

Why it matters for our MC

- Single RTL targets DDR3, DDR4, DDR5 and LPDDR4
- Swapping PHY = different memory standard, same controller